

Beyond One-Step Error: Free-Running Evaluation of ICU Chart-Process World Models Across Cohorts

Anonymous Authors

Affiliations withheld for anonymous review

Abstract—Background— Clinical world models are increasingly evaluated as sequence forecasters and simulators. One-step error does not establish that a model remains accurate, uncertainty-aware, constraint-consistent, or transportable when it generates its own future inputs. We evaluated models of recorded ICU values and measurement masks under capability-matched free-running assessment.

Methods— We used 40,336 intensive-care-unit patient stays from the PhysioNet/Computing in Cardiology Challenge 2019. Models were developed and selected in cohort A, then evaluated on an internal cohort-A test set and a frozen cohort-B test set. A separate cohort-B partition was used only for a post-pilot spread-scaling analysis. Last observation carried forward, population medians, ridge vector autoregression, a probabilistic GRU-D-style model, a causally masked Transformer, and a recurrent state-space model were compared on common patient-time anchors at 1, 3, 6, 12, and 24 hours. Outcomes included patient-macro normalized mean absolute error, continuous ranked probability score, a moment-matched Gaussian negative log score, empirical 90% interval coverage from 20 trajectories, observation-mask Brier score, and prespecified chart-state constraint violations. Neural results were repeated across five fixed seeds.

Results— Ridge vector autoregression had the lowest external one-hour error (NMAE 0.3610), but the GRU-D-style model and Transformer were better at 12 hours (0.5216 and 0.5207 versus 0.5301) and 24 hours (0.5607 and 0.5591 versus 0.5862). Their external 12-hour difference was 0.0008 (patient-bootstrap interval conditional on five fitted instances, -0.0000 to 0.0017). The state-space model was less accurate (12-hour NMAE 0.6220) and strongly underdispersed under both the 20-draw primary evaluation (coverage 0.4981) and a 100-draw sensitivity (0.5491). Spread scaling improved 20-draw empirical coverage without changing point forecasts, but coverage for GRU-D and the Transformer was materially trajectory-count dependent. The two most accurate long-horizon models also had the highest arterial-pressure ordering violation rates. All models degraded externally, although 12-hour rank order was stable (Spearman correlation 0.993).

Conclusions— One-step accuracy, long-horizon accuracy, empirical uncertainty, and narrow constraint diagnostics gave different model conclusions. Medical world-model evaluations should report these properties jointly, disclose Monte Carlo sensitivity, and separate spread adaptation from improved dynamics. These findings concern passive chart-process forecasting only.

Index Terms—world model, clinical time series, free-running rollout, uncertainty, external validation, intensive care

I. INTRODUCTION

A clinical dynamics model can perform well when predicting the next observed value yet fail when asked to generate a trajectory. During free-running rollout, each sampled state and measurement pattern becomes part of the next input. Small value, uncertainty, or observation-process errors can

therefore accumulate, change the effective state distribution, and produce trajectories that no longer resemble the data on which the model was trained [5].

This distinction matters for medical world models. Here, the modeled object is the recorded chart process: observed values, measurement masks, and carry-forward states under an existing process of care. It is not the latent patient physiology independent of treatment, measurement, discharge, or death. A one-step forecaster only needs to approximate the next conditional observation under a real history. A world model used for simulation must additionally remain stable under its own generated history, express uncertainty that corresponds to empirical error, preserve clinically meaningful state relationships, and reveal when those properties change across cohorts [6], [7], [8]. These requirements precede any stronger claim about treatment effects, counterfactual outcomes, or planning.

An AI-assisted scoping review, still pending qualified human dual verification, identified a candidate joint gap: free-running and horizon-resolved reporting were often present, while uncertainty assessment and frozen external evaluation were rarely combined. This preliminary finding motivated a capability-matched benchmark rather than a new architecture claim.

Adjacent work addresses different parts of the problem. GRU-D models informative missingness [3], and latent ordinary differential equations model irregular observation times and continuous latent dynamics [9]. Longitudinal EHR generators such as EHR-Safe and HALO emphasize synthetic-record fidelity, downstream utility, and privacy [10], [11]. Clinical external-validation studies also show that discrimination and calibration can change across hospitals [12]. These contributions do not by themselves answer whether the same ICU value-and-mask forecaster remains accurate, uncertainty-aware, and constraint-consistent during common-support free-running rollout.

We therefore developed MedWM-Eval-ICU, a real-data benchmark for passive clinical chart-process forecasting. The study asks whether model conclusions change when assessed by free-running error, empirical uncertainty, observation-process fidelity, prespecified constraint consistency, and cross-cohort transfer rather than one-step error alone.

Four estimation-focused research questions corresponded to the initial protocol:

1. how strongly do one-hour model rankings agree with 12- and 24-hour free-running rankings;

2. how do empirical interval coverage, probabilistic scores, and prespecified constraint violations change with rollout horizon;
3. how do performance and ranking change between the internal and zero-shot external cohorts; and
4. does observation-mask fidelity distinguish models that appear similar on value error.

An additional exploratory question asked whether an explicitly stochastic latent dynamics model was necessarily better dispersed than simpler probabilistic forecasters. This comparison was formalized after the first implementation seed and before the remaining seed matrix; it is not treated as a protocol-defined hypothesis. The protocol’s proposed scheduled-sampling comparison was not implemented in this study.

II. METHODS

A. Study design and claim boundary

This was a retrospective benchmark study using deidentified public data. The target capability was passive forecasting of future recorded values and measurement masks. The generated rollout is therefore a chart-process simulation under observed care, not a treatment-independent patient-state simulation. Sepsis labels were excluded from model inputs. The dataset does not provide a sufficiently complete treatment-action stream for causal intervention or policy evaluation.

The initial protocol, split logic, preprocessing, model families, horizons, and primary metrics were fixed before the final comparative analysis. A first implementation seed was used to verify the end-to-end neural pipeline. The common-support correction, five-seed matrix, and spread-scaling procedure were then frozen before the remaining seeds were run. Model development and hyperparameter selection used cohort A only. A dated amendment record distinguishes original, corrected, deferred, and exploratory analyses.

B. Data source and integrity audit

We used version 1.0.0 of the PhysioNet/Computing in Cardiology Challenge 2019 dataset (DOI 10.13026/v64v-d857), distributed under CC BY 4.0 [1], [2]. The source contains hourly ICU records from two public cohorts: 20,336 patients and 790,215 patient-hours in cohort A, and 20,000 patients and 761,995 patient-hours in cohort B.

The local integrity audit confirmed all 40,336 expected patient files, the expected schema, monotonically increasing ICU time, and stable patient-level fields. No malformed rows or invalid headers were detected. Aggregate fingerprints were stored without redistributing patient-level source files.

C. Patient-level partitions

Cohort A was divided deterministically into 14,247 training, 3,015 validation, and 3,074 internal-test patients. Cohort B was divided into 2,069 external-calibration and 17,931 external-test patients. No patient occurred in more than one partition.

The external-test partition was not used for preprocessing, model fitting, early stopping, hyperparameter selection, or calibration. Zero-shot cohort-B results and results after external-calibration spread scaling are reported as separate analyses.

D. Variables and preprocessing

The candidate feature set comprised 34 hourly physiological variables. A variable was retained when its observation frequency was at least 5% in the cohort-A training partition. This yielded 24 variables: heart rate, oxygen saturation, temperature, systolic, mean, and diastolic blood pressure, respiratory rate, base excess, bicarbonate, inspired oxygen fraction, pH, arterial carbon dioxide, arterial oxygen saturation, blood urea nitrogen, chloride, creatinine, glucose, magnesium, phosphate, potassium, hematocrit, hemoglobin, white blood-cell count, and platelets.

All medians, interquartile ranges, coverage decisions, and clipping statistics were estimated from cohort-A training patients. Observed values were robustly scaled. Missing values in model inputs were carried forward after a training-median initialization and were accompanied by an observation mask and hours since last measurement. Elapsed time since measurement was clipped at 24 hours and scaled to [0,1]. The ICU-time feature, rather than the patient record, was capped at 72 hours and scaled to [0,1]. Imputed values were never scored as observed targets.

E. Forecasting task and common support

For each eligible anchor, a model received up to 72 hours of history, with a minimum context of 12 hours, and generated future values and measurement masks at 1, 3, 6, 12, and 24 hours. The primary evaluation used free-running rollout: sampled values and masks were recursively returned to the model rather than replaced by future observations. The one-hour endpoint is a single generated step from the real history. The protocol’s proposed teacher-forced comparison at longer horizons was deferred and is not represented as completed.

Every reported horizon used the same anchors. An anchor was eligible only when all 24 future hours were available, and anchors were sampled every six hours. Because value metrics exclude missing targets, the number of scored targets and patients with at least one scored target still varies by horizon. Common support therefore denotes identical anchor eligibility, not identical horizon-specific observed-target support.

F. Models

We evaluated three deterministic sanity baselines:

- last observation carried forward;
- a training-cohort global median; and
- a training-cohort median conditional on ICU hour.

A ridge vector autoregression used three lagged robust-scaled states and an intercept. Its regularization coefficient was selected by validation-set 12-hour patient-macro normalized mean absolute error and was then frozen.

The probabilistic neural comparison comprised:

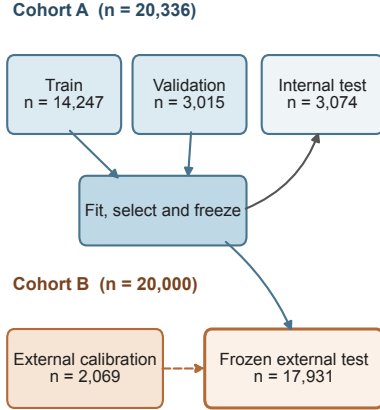
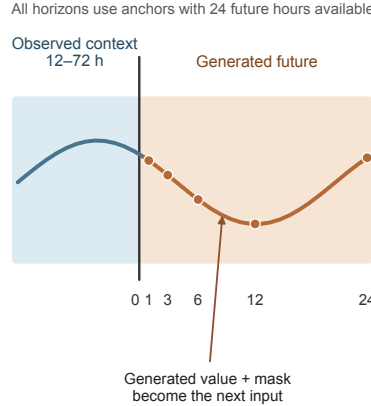
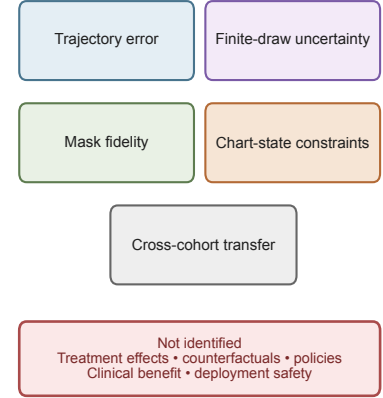
a Patient-level splits and evidence path**b Free-running rollout on common support****c Measured evidence and claim boundary**

Fig. 1. Evaluation design, free-running rollout, and claim boundary. Panel (a) shows the patient-disjoint development, internal-test, external-calibration, and frozen external-test partitions. The external-calibration partition can adjust predictive spread but cannot alter model parameters or select a model. Panel (b) shows the common-support free-running task: every reported horizon uses anchors with all 24 future hours available, and generated values and measurement masks become later inputs. Panel (c) distinguishes the evidence measured from claims that the observational passive-forecasting design does not identify.

- a probabilistic GRU-D-style recurrent forecaster using values, masks, and time since measurement [3];
- a causally masked Transformer forecaster with cached autoregressive inference [4]; and
- a recurrent stochastic state-space model with a 32-dimensional latent state, adapted from latent-dynamics modeling principles [5].

All neural models used a hidden size of 96, a Gaussian value head, and a Bernoulli measurement-mask head. The Transformer used three layers, four attention heads, and dropout 0.1; the recurrent models did not use an explicit dropout layer. The resulting parameter counts differed by architecture and are reported with the results. Models were optimized with AdamW at learning rate 0.0003, weight decay 0.00001, gradient clipping at 1.0, and batch size 64. Training was limited to 20 epochs with validation early stopping. The objective combined observed-value Gaussian negative log likelihood with mask cross-entropy; the state-space model additionally used a weighted latent Kullback-Leibler term. Training and validation objectives were averaged over minibatches; publication outcomes use the separate patient-macro aggregation defined below. Complete window sampling, loss weights, scale bounds, checkpoint selection, and parameter counts are reported in Supplementary Table S4.

Evaluation used batches of 128 patient-time anchors to reduce accelerator kernel overhead; this setting did not change model parameters, support, or metric definitions.

Each neural family was trained with seeds 20260912 through 20260916 under the same data, maximum epoch count, stopping rule, optimizer settings, and evaluation code. Architecture-specific parameter counts were not matched, so

comparisons are fixed-budget benchmarks of these implementations rather than controlled mechanistic ablations or population-level model-family rankings. All GRU-D and Transformer runs selected the maximum epoch 20, so longer training could change their results. Ridge and neural models also used different validation-selection objectives. Accelerator kernels may remain nondeterministic, and seed-level results are retained rather than represented as exact computational replicates.

G. Probabilistic rollout and post-hoc spread adaptation

Twenty stochastic trajectories were generated for each patient-time anchor. Point forecasts were the trajectory mean. Distributional performance was evaluated from the 20-draw empirical trajectory distribution. The reported 90% interval used linearly interpolated empirical 5th and 95th percentiles. With 20 draws, those quantiles, the sample mean, CRPS, and the sample-standard-deviation score include non-negligible Monte Carlo error and do not directly characterize the infinite-draw model distribution. Under exchangeable draws from a calibrated continuous distribution, the implemented linearly interpolated interval has expected finite-sample coverage of approximately 0.814 at 20 draws, 0.865 at 50 draws, and 0.882 at 100 draws, rather than exactly 0.90.

The initial protocol anticipated a separate external-calibration analysis. The exact horizon-wise scalar procedure was fixed after the implementation seed and before the remaining seed matrix, so it is treated as a post-pilot adaptation analysis. One nonnegative spread multiplier was fitted per horizon on the external-calibration partition. For raw predictive

samples with mean μ , standard deviation σ , and observed target y , the multiplier was

$$s_h = \sqrt{\text{mean} [((y - \mu)/\sigma)^2]}$$

over observed targets at horizon h . Spread-adapted samples were $\mu + s_h(x - \mu)$. This operation changed marginal spread only. It did not change the point forecast, generated rollout states, predicted measurement masks, or raw trajectory constraint counts.

The multiplier was fitted from the same 20-draw procedure by pooling observed targets, whereas test outcomes were aggregated patient first. It therefore targets pooled marginal spread, not variable-specific, patient-conditional, joint, or measurement-mask calibration. It can compensate for both model dispersion and finite-draw estimation and is not interpreted as a pure correction of the underlying model distribution.

H. Outcomes

The primary descriptive outcomes were:

1. patient-macro normalized mean absolute error at 12 hours;
2. patient-macro continuous ranked probability score at 12 hours;
3. area under the 1- to 24-hour rollout-degradation curve;
4. coverage of the empirical 5th- to 95th-percentile interval, with 0.90 retained only as an infinite-draw reference;
5. prespecified constraint violations per 1,000 sampled rollout states;
6. relative degradation from internal to external testing; and
7. patient-macro measurement-mask Brier score.

NMAE was absolute error after scaling each variable by the cohort-A training interquartile range, averaged across scored variable-anchor targets within a patient and then equally across patients. Secondary outcomes included one-hour error, a moment-matched Gaussian negative log score, interval width, and variable-specific metrics. The negative log score used the empirical trajectory mean and standard deviation and is not the exact likelihood of a generally non-Gaussian rollout distribution. CRPS was used as a proper scoring rule for the empirical predictive distribution [6]; interval coverage and spread adaptation were interpreted as finite-draw uncertainty diagnostics rather than substitutes for accuracy [7], [8].

Prespecified chart-state checks included systolic pressure greater than or equal to mean pressure greater than or equal to diastolic pressure, oxygen saturation in $[0,100]$, arterial oxygen saturation in $[0,100]$, and inspired oxygen fraction in $[0,1]$. The primary constraint metric used the actual rollout state after applying the sampled measurement mask and carrying unmeasured values forward. Violations in every decoder emission were retained as a separate diagnostic. Because masks can update pressure components at different simulated times, a rollout-state ordering violation may reflect asynchronous carry-forward inconsistency rather than an impossible simultaneous measurement. No matched observed-data reference rate was

estimated. Both diagnostics were calculated from the raw trajectory so that spread scaling could not alter them.

I. Statistical analysis

All comparisons use identical patient-time anchors within a split. Metrics are aggregated across anchors and observed variables within each patient and then equally across patients. Neural patient metrics are averaged over the five fixed trained instances; across-seed standard deviation and range are reported separately. The primary estimand therefore concerns these five instances, not the full distribution of possible training runs.

Patient-bootstrap intervals resample patients after fixed-seed averaging. Paired bootstrap contrasts are restricted to the frozen 12-hour NMAE comparisons on the same patients. Internal and external cohorts contain different patients, so cross-cohort intervals resample each cohort independently. Rank stability is described by Spearman correlation and the fraction of model pairs whose order reverses across horizons or cohorts. Patient-bootstrap intervals quantify patient sampling conditional on the five fitted neural instances; they do not include training-run uncertainty. Across-seed standard deviations describe those instances separately.

For horizon-level metric M_h , the normalized rollout area is the trapezoidal area across 1, 3, 6, 12, and 24 hours divided by 23 hours. External degradation is reported as $M_B - M_A$ and $100(M_B - M_A)/M_A$, where A is internal and B is zero-shot external.

The initial protocol proposed multiplicity control but did not define exact null contrasts. We therefore emphasize effect sizes and 95% bootstrap intervals and do not present retrospectively specified p -values as confirmatory tests. Analyses formalized after the implementation seed are labeled exploratory or post-pilot. A sensitivity analysis excludes the inspected implementation seed.

Because empirical quantiles and sample-based proper scores can depend on Monte Carlo count, a post-analysis numerical sensitivity reran the implementation seed with 20, 50, and 100 trajectories on the same hashed subset of 1,000 external-test patients. This analysis assessed estimator stability rather than training-seed uncertainty and did not replace the prespecified five-seed, 20-trajectory primary analysis. The three counts used separate fixed random streams rather than a repeated-stream convergence design.

Missing targets do not enter value metrics. Mask metrics use all target variables. No test-set result is used to select a model or adjust a hyperparameter.

J. Reproducibility

The local reproducibility package records the dataset fingerprint, deterministic patient split, selected variables, training configuration, model checkpoints, seed-level training histories, raw patient-level metrics, spread-adaptation artifacts, and scripts used to generate summary tables. Source patient files are not redistributed. An immutable reviewer-accessible release remains a submission gate.

III. RESULTS

A. Cohort and evaluation support

After applying the 12-hour minimum context and 24-hour common-support rule, 1,841 internal-test and 10,622 external-test patients contributed at least one eligible anchor, yielding 5,143 and 30,450 common anchors, respectively. Across the 24 retained variables, the observed-value fraction was 28.36% in the full internal-test records and 26.04% in the full external-test records. Effective value-metric support varied with observed targets: internal support ranged from 1,810 patients and 36,216 targets at one hour to 1,773 and 32,924 at 24 hours; external support ranged from 10,245 patients and 198,911 targets to 10,162 and 192,680. The 12-hour paired comparisons used 1,806 internal and 10,178 external patients.

B. One-step and free-running accuracy

The one-hour and long-horizon comparisons did not select the same model (Figure 2). Externally, ridge had the lowest one-hour NMAE (0.3610), followed by the Transformer (0.3768), GRU-D (0.3770), and persistence (0.3853). At 12 hours, the Transformer (0.5207) and GRU-D (0.5216) were better than ridge (0.5301); the same ordering remained at 24 hours (0.5591, 0.5607, and 0.5862). The state-space model reached NMAE 0.6220 at 12 hours.

At 12 hours, the external paired contrast was -0.0093 (95% patient-bootstrap interval conditional on the five fitted instances, -0.0114 to -0.0074) for Transformer versus ridge and -0.0085 (-0.0102 to -0.0068) for GRU-D versus ridge. GRU-D minus Transformer was 0.0008 (-0.0000 to 0.0017), compatible with essentially tied performance. Ridge was better than persistence by -0.0452 (-0.0488 to -0.0414), while the state-space model was worse than ridge by 0.0919 (0.0896 to 0.0942).

The prespecified external NMAE rollout-area summary was 0.5083 (across-seed SD 0.0038) for the Transformer, 0.5087 (0.0016) for GRU-D, 0.5158 for ridge, 0.5516 for persistence, 0.6105 (0.0054) for the state-space model, 0.6652 for the hourly median, and 0.6656 for the global median. Excluding the inspected implementation seed did not change the main pattern: 12-hour NMAE was 0.5213 for the Transformer, 0.5216 for GRU-D, and 0.6234 for the state-space model.

Original-unit 12-hour errors were heterogeneous across variables (Supplementary Table S1). The Transformer versus ridge MAE was 9.65 versus 9.84 for heart rate, 1.96 versus 2.03 for oxygen saturation, 15.28 versus 16.74 for systolic pressure, and 10.33 versus 11.38 for mean pressure. Ridge was better for temperature (0.502 versus 0.511), creatinine (0.403 versus 0.442), glucose (29.55 versus 30.87), and white-cell count (2.58 versus 2.88). The aggregate NMAE advantage was therefore not uniform across clinical variables, and no minimum clinically important difference was defined.

C. Finite-draw uncertainty and spread adaptation

Under the 20-draw primary procedure, 12-hour external 90% coverage was 0.8422 for GRU-D, 0.8673 for the Transformer, and 0.4981 for the state-space model (Figure 3). Spread

adaptation increased coverage to 0.8841, 0.8866, and 0.8784, respectively. These changes describe the implemented empirical interval and are not interpreted as reductions in intrinsic model-calibration error.

Mean interval width increased from 1.9116 to 2.1839 scaled units for GRU-D, 2.1120 to 2.2555 for the Transformer, and 1.0269 to 2.6629 for the state-space model. Gaussian negative log score improved from 1.1100 to 1.0874, 1.0927 to 1.0886, and 3.2856 to 1.2795. CRPS became slightly worse for GRU-D (0.3850 to 0.3885) and the Transformer (0.3851 to 0.3874) but improved for the state-space model (0.4957 to 0.4706). Spread adaptation therefore altered the 20-draw empirical distribution without making the state-space model competitive on accuracy. The maximum patient-level raw-versus-adapted NMAE difference was 4.768×10^{-7} , and mask-Brier difference was zero.

D. Observation-process fidelity

At 12 hours, external mask Brier scores were 0.0617 for GRU-D, 0.0628 for the Transformer, and 0.0645 for the state-space model; internal values were 0.0730, 0.0725, and 0.0797. The lower external scores occurred with lower measurement prevalence and should not be read as improved patient-state transport. At 12 hours within each cohort, the state-space model had both the weakest value forecast and the weakest observation-process forecast.

E. Prespecified chart-state constraint diagnostics

Accuracy and narrow constraint consistency gave different rankings (Figure 4). At 12 hours, GRU-D and the Transformer produced 161.1 and 204.9 pressure-order violations per 1,000 rollout states, compared with 2.50 for the state-space model, 0.624 for ridge, 5.32 for persistence, and zero for median baselines. For GRU-D, Transformer, and the state-space model, respectively, rates were 123.0, 158.8, and 11.2 for oxygen-saturation bounds; 50.8, 49.4, and 70.3 for arterial-oxygen-saturation bounds; and 5.7, 8.3, and 0.2 for inspired-oxygen fraction.

Low violation counts did not imply useful dynamics: the state-space model had worse NMAE and uncertainty, and median forecasts satisfied pressure ordering by construction. Conversely, the two most accurate long-horizon models often violated the pressure-order rule. Because mask-gated values can have different carry-forward ages and no matched observed-chart reference was computed, these rates are descriptive consistency diagnostics, not estimates of impossible simultaneous physiology or clinical safety.

F. Cross-cohort degradation and ranking stability

Every model had higher 12-hour NMAE externally. Relative degradation was 3.9% (95% interval, 2.0% to 5.7%) for ridge, 4.9% (3.1% to 6.9%) for GRU-D, 5.2% (3.3% to 7.1%) for the Transformer, 4.7% (2.7% to 6.6%) for the state-space model, and 4.1% (2.1% to 6.4%) for persistence.

Cross-cohort rank order among seven models remained stable: mean Spearman correlation was 0.993 at 12 hours and

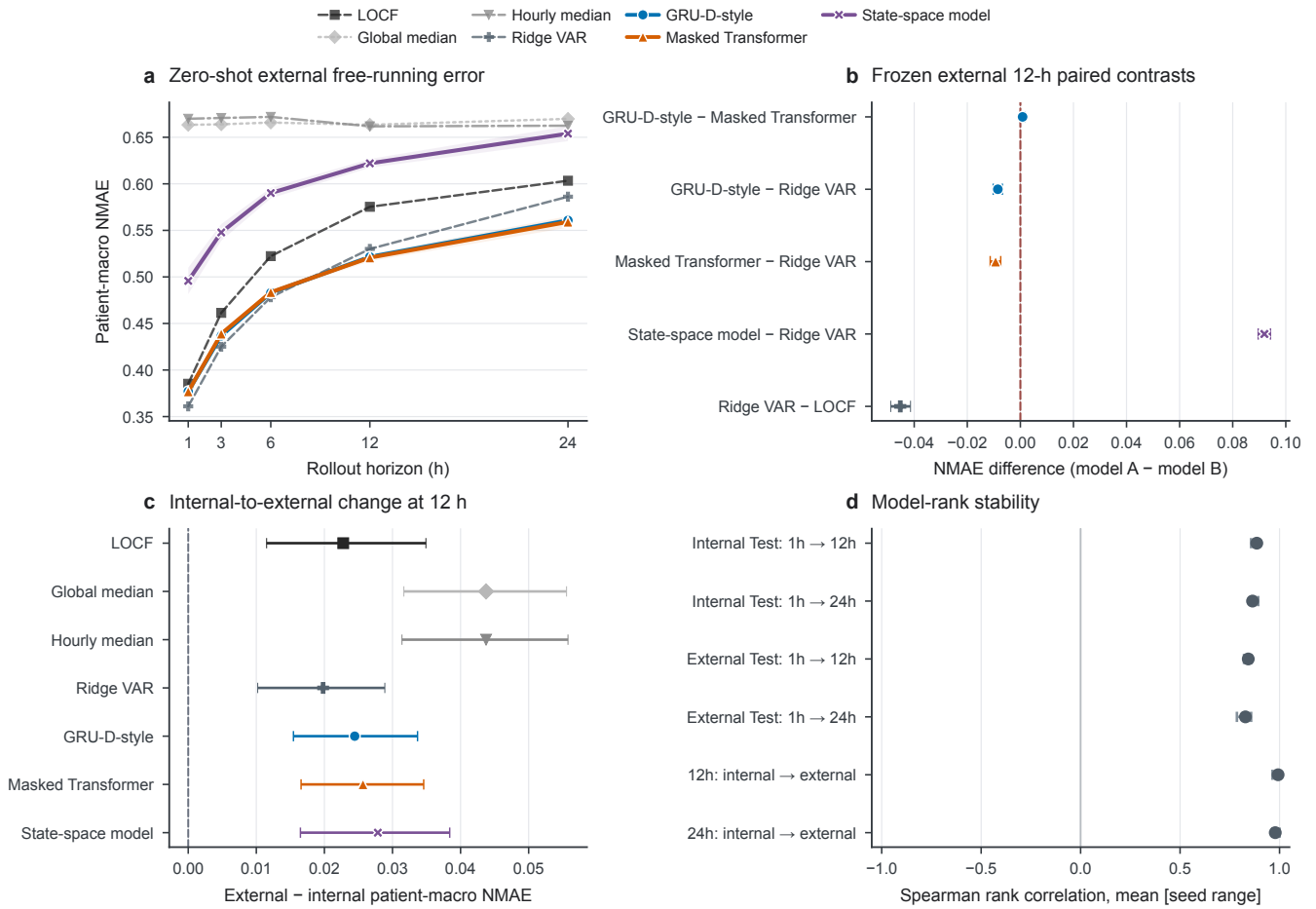


Fig. 2. Free-running accuracy and transfer. (a) Zero-shot external NMAE by horizon; ribbons show one across-seed SD for neural models. (b) Frozen external 12-hour patient-bootstrap contrasts conditional on five fitted instances; negative values favor model A. (c) Internal-to-external 12-hour change with independently resampled patient-bootstrap intervals. (d) Model-rank correlations among seven models; bars show seed ranges.

0.979 at 24 hours. Horizon transfer was less complete. External one-to-12-hour correlation was 0.843, with 16.2% of model pairs reversing; one-to-24-hour values were 0.829 and 17.1%.

Trajectory-count sensitivity qualified raw uncertainty estimates. On the same 1,000 external patients and implementation seed, 12-hour coverage increased from 0.8401 with 20 trajectories to 0.8945 with 100 for GRU-D and from 0.8632 to 0.9195 for the Transformer. Their CRPS and sample-mean NMAE also decreased. The state-space model increased only from 0.4978 to 0.5491 and remained underdispersed. Pressure-order rates were comparatively stable.

G. Descriptive zero-shot external-test result table

Baseline values use external-test patient-macro means on the common anchor set. Neural entries are mean (across-seed standard deviation). The displayed constraint rate is the 12-hour arterial-pressure ordering rate.

IV. DISCUSSION

This study yields four main findings. First, one-step accuracy did not fully determine free-running performance. Ridge was best at one hour, but GRU-D and the Transformer were

better at 12 and 24 hours and essentially tied externally. Long-horizon rollout changed the conclusion without establishing an architectural winner.

Second, stochastic architecture did not guarantee useful uncertainty. The state-space model contained latent randomness yet produced narrow intervals, approximately 50% 12-hour coverage, and the worst CRPS and NMAE. Spread scaling changed finite-draw coverage and negative log score but not the learned mean trajectory. For all models, the 20-draw multiplier combines model spread with Monte Carlo estimation and cannot be interpreted as a pure calibration correction.

Third, the most accurate neural models had the highest pressure-order violation rates in mask-gated chart states. Favorable average error can therefore coexist with failure of a narrow prespecified consistency rule. The converse also failed: state-space and median models had fewer violations without useful overall accuracy. These checks must be reported beside error, not used as a validity or safety certificate. Constrained decoding is a plausible next intervention, but its effects require direct testing.

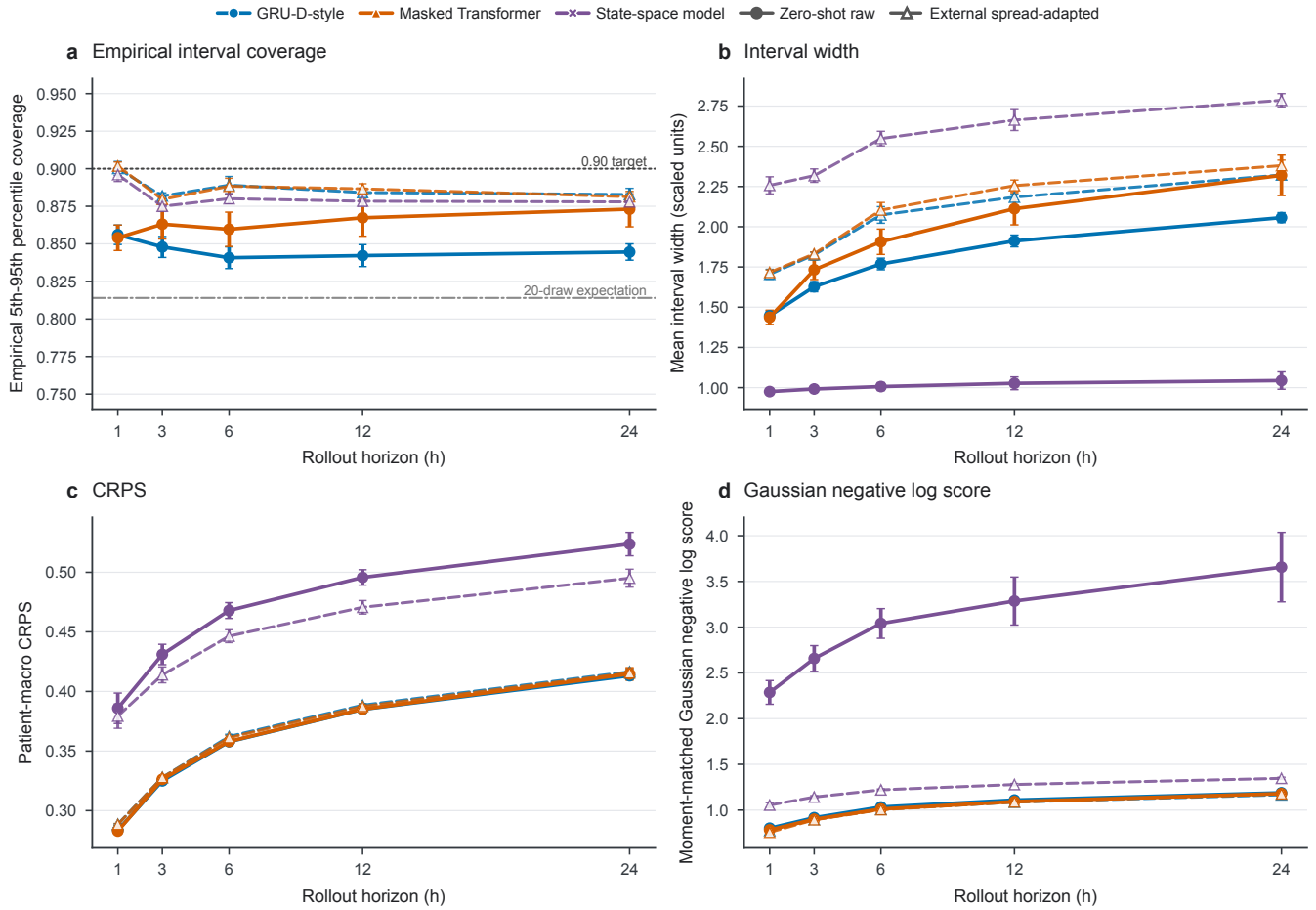


Fig. 3. External 20-draw uncertainty before and after spread adaptation. Panels show (a) empirical 5th- to 95th-percentile coverage, (b) interval width, (c) CRPS, and (d) moment-matched Gaussian negative log score. Circles are zero-shot results and triangles are horizon-specific spread scaling fitted on external-calibration data with the same 20-draw procedure. Error bars show across-seed SDs, not patient-bootstrap intervals. The dotted line is the infinite-draw 0.90 target; the dash-dotted line is the approximately 0.814 expectation for the implemented 20-draw quantile rule under exchangeability. These panels characterize the finite-draw procedure rather than the infinite-draw model distribution.

TABLE I
DESCRIPTIVE ZERO-SHOT EXTERNAL-TEST RESULTS. NEURAL VALUES ARE MEAN (ACROSS-SEED STANDARD DEVIATION).

Model	1 h NMAE	12 h NMAE	24 h NMAE	12 h CRPS	Empirical coverage	Mask Brier	Pressure violations/1,000
Last observation carried forward	0.3853	0.5753	0.6033	NA	NA	NA	5.320
Global median	0.6634	0.6634	0.6698	NA	NA	NA	0.000
Hourly median	0.6699	0.6617	0.6624	NA	NA	NA	0.000
Ridge vector autoregression	0.3610	0.5301	0.5862	NA	NA	NA	0.624
Probabilistic GRU-D-style model	0.3770 (0.0010)	0.5216 (0.0013)	0.5607 (0.0030)	0.3850 (0.0008)	0.8422 (0.0073)	0.0617 (0.0020)	161.067 (6.562)
Causally masked Transformer	0.3768 (0.0016)	0.5207 (0.0038)	0.5591 (0.0062)	0.3851 (0.0031)	0.8673 (0.0123)	0.0628 (0.0011)	204.920 (23.372)
Recurrent state-space model	0.4958 (0.0142)	0.6220 (0.0047)	0.6539 (0.0081)	0.4957 (0.0065)	0.4981 (0.0149)	0.0645 (0.0004)	2.496 (1.254)

Fourth, cohort shift increased error but barely changed rankings. This negative finding narrows the original hypothesis: here, horizon shift mattered more than cohort shift for comparative order. Stable ranking does not make external error acceptable, so frozen external testing remains necessary.

The trajectory-count sensitivity changes the interpretation of raw coverage. With 20 samples, empirical 5th and 95th percentiles are not a finite-sample 90% prediction interval, and sample means, CRPS, and Gaussian scores retain simulation error. At 100 samples, GRU-D approached nominal

coverage and the Transformer moved slightly above it on the tested subset. The state-space model remained far below nominal. Because the sensitivity used one trained seed and separate single streams, it demonstrates dependence rather than convergence. Future evaluations should use finite-sample-valid intervals, repeated nested streams, or analytic summaries where available.

Together, the results support an evaluation bundle rather than a new architecture claim: common-eligibility free-running curves, proper scores, finite-draw uncertainty with Monte

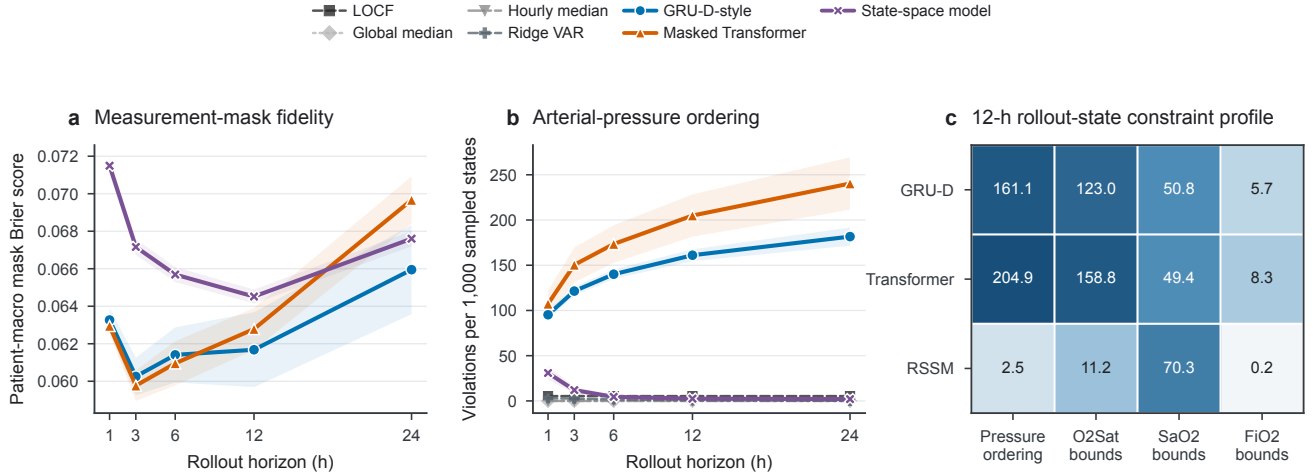


Fig. 4. Observation-process and chart-state constraint diagnostics. (a) External mask Brier score. (b) Arterial-pressure ordering violations in actual rollout states. (c) Four prespecified 12-hour constraints for neural models. Rates are per 1,000 sampled states; heat-map color encodes frequency, not clinical severity. Ribbons in panels (a) and (b) show one across-seed SD.

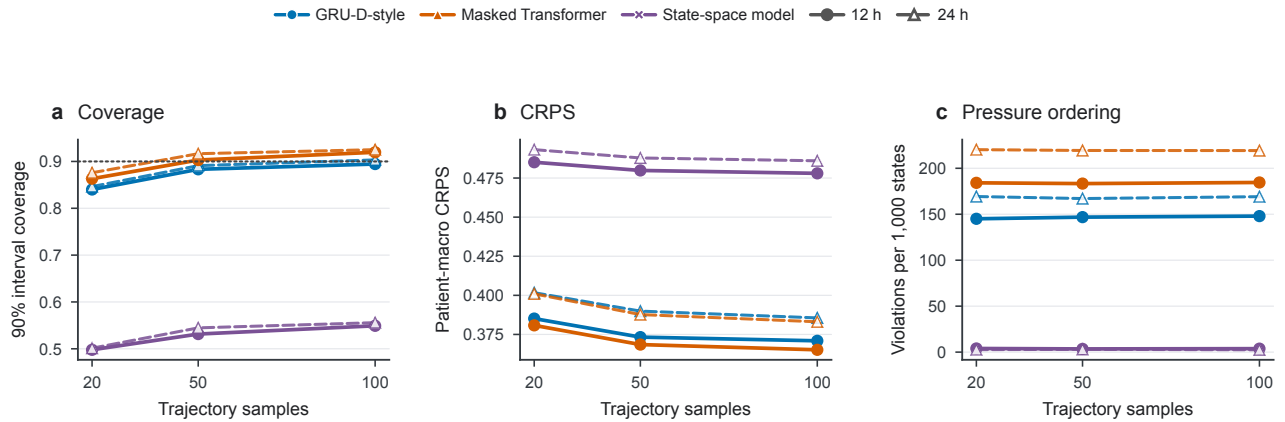


Fig. 5. Limited Monte Carlo-count sensitivity. Coverage, CRPS, and pressure-order violations for 20, 50, and 100 trajectories on the same hashed subset of 1,000 external patients. This analysis uses one trained implementation seed, separate fixed streams, and no repeated-stream Monte Carlo intervals. Circles and triangles denote 12- and 24-hour horizons.

Carlo sensitivity, observation-process fidelity, prespecified chart-state constraints, and frozen external testing. This bundle assesses a simulator of the recorded passive chart process; it does not establish latent physiological, causal, or clinical validity.

A. Limitations

The study uses one challenge dataset with two source cohorts rather than prospective deployment data. Cohort B shares the challenge schema, so it tests transfer but not general deployment validity. Sparse observation limits variable-specific precision. The 20-trajectory primary analysis mixes model behavior with Monte Carlo error, as the sensitivity

analysis demonstrates. Spread adaptation is pooled and scalar by horizon, not conditional or joint. Constraint rates lack a matched observed-chart reference, patient-clustered intervals, and clinician validation; pressure components may have different carry-forward ages. Variable-level original-unit errors were descriptive and lacked clinician-validated importance thresholds, so small aggregate NMAE differences have no defined minimum clinically important interpretation. Models omit static variables, treatment actions, discharge, and death. Parameter counts differ, hyperparameter exploration was limited, GRU-D and Transformer training ended at the epoch cap, and architecture differences cannot be attributed to one mechanism.

V. CONCLUSION

One-step error, long-horizon accuracy, finite-draw uncertainty, and narrow chart-state constraints did not identify the same implementation. Ridge was strongest at one hour; GRU-D and the Transformer were strongest at 12 and 24 hours but had high pressure-order violation rates; and the stochastic state-space model was inaccurate and robustly underdispersed. Spread scaling changed the 20-draw empirical distribution without repairing dynamics, while cohort shift raised errors but left rankings mostly stable. Medical world-model claims should therefore be matched to free-running, uncertainty, constraint, and transfer evidence reported together. The present evidence is limited to passive chart-process forecasting.

VI. DATA AND CODE AVAILABILITY

The source dataset is publicly available from PhysioNet under its published terms. The local reproducibility package contains code, split hashes, aggregate data fingerprints, trained checkpoints, spread-adaptation artifacts, patient-level derived metrics, source data for figures, and result tables; it does not redistribute source patient files. A stable immutable release is required before submission.

VII. ETHICS

This study uses deidentified public data and performs retrospective model evaluation only. The research team must confirm the applicable institutional determination before submission.

VIII. FUNDING

Funding information is withheld for anonymous review.

IX. DECLARATION OF COMPETING INTEREST

Competing-interest information is withheld for anonymous review.

X. DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

OpenAI Codex was used for code assistance, evidence organization, drafting, and language revision under author supervision. The authors will verify all methods, results, citations, and interpretations and take full responsibility for the final manuscript.

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